

Half / full adder

Addition starts with a half adder

Sum, carry, chain

- Half adder truth
- Full adder with A, B, and C_{in} all one gives sum one and carry-out one, majority wins
- Compose view
- Ripple two bits

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

Half / full adder builder

Wire the classic blocks: **half adder** → **full adder** (two HAs + OR) → a tiny **ripple** chain.

Starter example: half adder with A=1, B=1 → S=0, C=1 (binary 1+1). Then promote to a full adder / two-HA build.

Load starter example

Challenges (1/22)

Quiz: HA sum. Half-adder sum is which gate of A,B? Answer: XOR

Your answer

Show hintCheckNextIdle

quiz-ha-squiz-ha-cquiz-fa-inquiz-ha-limitstarter-haha-01preset-fafa-majoritycompose-modecompose-orquiz-composeripple-presetripple-2+1explainquiz-xor3togglemode-faquiz-cin0ha-eq-faquiz-ripplecompose-s1full-path

Core ideas

Half adder
S = A⊕B, C = A·B — no Cin.

Full adder
S = A⊕B⊕Cin, Cout = majority.

Build
FA = HA(A,B) + HA(S₁,Cin) + OR carries.

Half full adder starter

learn_digital — half full adder

Workbook practice

- In the workbook track, fill the half-adder truth table for all four A-B pairs
- For full adder A equals one, B equals zero, Cin equals one, give S and Cout
- Sketch two half adders and an OR as one full adder
- For two-bit ripple with A equals one-zero and B equals one-one, name S zero, S one
- Name one pitfall: using a half adder where a column needs carry-in

Pitfalls to watch

- Do not confuse half-adder carry with full-adder carry-out, they solve different problems
- Majority carry is not the same as HA carry alone when C_{in} is in play
- And remember: the browser lab is literacy
- Real adders still need timing, overflow flags

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one HA truth table and one FA from two HAs
- When you are ready, take the short quiz, then continue to XOR parity tree